



S1.4 Counting particles by mass: the mole

Structure 1 · Models of the particulate nature of matter · IB Diploma · Concept guide and original exam-style practice

Understand the concept

The mole is the SI unit for amount of substance, defined so that one mole contains exactly Avogadro's number (6.02×10^{23}) of particles. Molar mass (M , g mol^{-1}) links mass to amount of substance via $n = m/M$, allowing chemists to convert between mass, moles and number of particles. Empirical and molecular formulas are determined from percentage composition or combustion data by converting masses to moles and finding the simplest whole-number ratio; molar concentration ($c = n/V$) extends mole calculations to solutions.

Worked example: follow the reasoning

A compound contains 40.0% carbon, 6.7% hydrogen and 53.3% oxygen by mass. Determine its empirical formula. (Ar: C = 12.0, H = 1.0, O = 16.0)

Step 1: Assume 100 g sample: 40.0 g C, 6.7 g H, 53.3 g O.

Step 2: Convert to moles: $n(\text{C}) = 40.0/12.0 = 3.33 \text{ mol}$; $n(\text{H}) = 6.7/1.0 = 6.7 \text{ mol}$; $n(\text{O}) = 53.3/16.0 = 3.33 \text{ mol}$.

Step 3: Divide by the smallest value (3.33): C = 1.00, H = 2.01, O = 1.00.

Answer: Empirical formula = CH_2O .

Quick check · multiple choice

What amount (in mol) of oxygen atoms is present in 0.50 mol of $\text{Al}_2(\text{SO}_4)_3$?

- A 0.50 mol
- B 3.0 mol
- C 6.0 mol
- D 12 mol

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A hydrated salt has the formula $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$. A 6.15 g sample of the hydrated salt is heated to constant mass, leaving 3.00 g of anhydrous MgSO_4 . [5]
(Molar masses: $\text{MgSO}_4 = 120.4 \text{ g mol}^{-1}$, $\text{H}_2\text{O} = 18.0 \text{ g mol}^{-1}$)
(a) Calculate the mass of water lost on heating. [1]
(b) Calculate the amount, in mol, of MgSO_4 and of H_2O in the sample. [2]
(c) Determine the value of x and hence the formula of the hydrated salt. [2]

Final answer: _____



Answers and explanation

Question 1 [5]

(a) Mass of water = $6.15 - 3.00 = 3.15$ g. [1]

(b) $n(\text{MgSO}_4) = 3.00 / 120.4 = 0.0249$ mol; $n(\text{H}_2\text{O}) = 3.15 / 18.0 = 0.175$ mol. [2]

(c) $x = n(\text{H}_2\text{O})/n(\text{MgSO}_4) = 0.175 / 0.0249 = 7.03 \approx 7$. Formula: $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$. [2]

Quick check answer

C. Each formula unit contains 12 O atoms ($3 \times \text{SO}_4$, 4 O each), so $0.50 \text{ mol} \times 12 = 6.0 \text{ mol O}$ atoms.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Chemistry HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.